BRIDGE: Balancing Representations by Identifying and Generating Underrepresented Data

Anonymous Author(s)
Affiliation
Address
email

Abstract

Self-supervised learning inherits the coverage biases of its source data. When an unlabeled corpus is long-tailed or otherwise skewed, representation space becomes dense around common modes and thin where semantic coverage is weak. Loss-level fixes can improve optimization, but they do not directly repair this geometry. We introduce BRIDGE, a data-centric self-supervised training loop that alternates between representation learning, mean kNN sparsity scoring, and targeted image-to-image diffusion augmentation. Rather than chasing the extreme OOD tail, BRIDGE repairs the modal sparse band of the OOD-distance distribution, which is more likely to contain underrepresented but still recoverable examples. Across five source regimes, including long-tailed ImageNet-100, CIFAR-10, CIFAR-100, PASS, and the fully synthetic DiffusionDB, BRIDGE achieves the best average linear-probe and kNN transfer in every regime we study. Relative to source-matched SimCLR, it improves seven-task linear-probe averages by 2.41 to 4.89 points. Ablations show that the strongest configuration combines SimCLR within the repair loop, Stable Diffusion 3 for generation, and the new mode-window selector instead of the older top-tail rule.

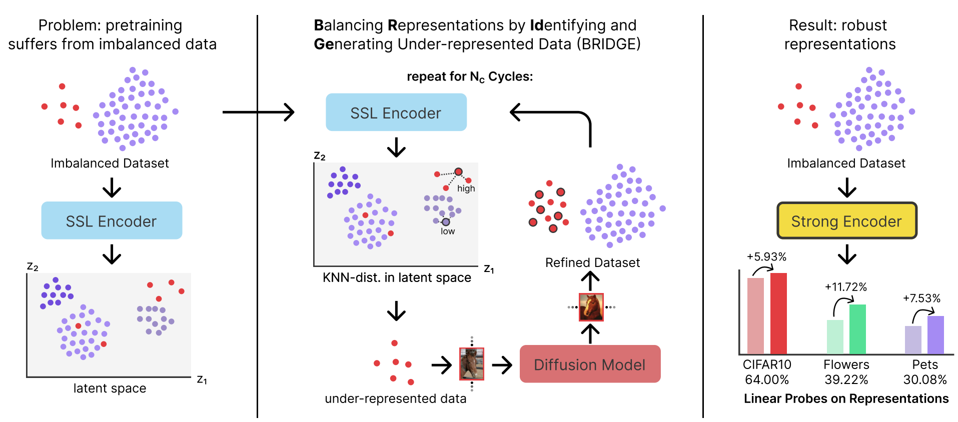

Figure 1: Conceptual overview of BRIDGE. We alternate between SSL pretraining, sparsity scoring in representation space, and targeted image-to-image diffusion augmentation. The final method augments the modal sparse band of the OOD-distance distribution rather than the extreme tail.

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1 Introduction

Self-supervised learning is now a standard way to learn transferable visual representations from unlabeled images. Strong methods span contrastive learning, masked prediction, and self-distillation [Chen et al., 2020, He et al., 2020, 2022, Caron et al., 2021, Oquab et al., 2024]. Yet these methods still inherit whatever the source corpus underrepresents. When source data are long-tailed or otherwise skewed, learned features concentrate around frequent modes and become sparse in regions with poor coverage. The consequence is not only an optimization issue. It is also a data-coverage issue: some semantic regions of the source manifold remain weakly modeled, and downstream transfer degrades.

Most prior attempts to address long-tailed SSL are model-centric. Temperature Schedules reshapes the optimization trajectory of contrastive learning over time [Kukleva et al., 2023]. SDCLR introduces a self-competitor to emphasize harder and rarer samples [Jiang et al., 2021]. Other work adds external OOD data or frequency-aware training heuristics to the SSL loop [Bai et al., 2023, Lin et al., 2023]. These methods are strong baselines, but they still treat the source distribution as largely fixed.

We study the complementary direction and ask whether we can repair the source data while SSL training proceeds. We propose BRIDGE, a cyclic data-repair loop shown in Figure 1. After each pretraining stage, we embed the current training set, score each image by mean kNN distance, select a sparse but still semantically recoverable region of representation space, and generate new training images for that region with a frozen diffusion model. The next cycle then trains on the repaired dataset. In effect, the encoder reveals where the source manifold is thin, and the generator densifies those regions before the next round of learning.

Two parts of the final design are central. First, we no longer select the most extreme OOD samples. Instead, we build a histogram over OOD distances and allocate a fixed budget around the modal sparse band of that histogram. This avoids the super-OOD tail, which often contains noisy or weakly useful images, while retaining the underrepresented region that is most amenable to repair. We keep the pure tail rule as an explicit ablation. Second, we broaden the empirical scope well beyond the earlier label-derived long-tailed sources. In addition to ImageNet-100-LT, CIFAR-10-LT, and CIFAR-100-LT, we evaluate web-source unlabeled data drawn from PASS and DiffusionDB. Across these five regimes, BRIDGE is best on the seven-task average and remains strong under both linear-probe and kNN transfer.

We make three contributions. First, we introduce a data-centric SSL loop that repairs sparse representation regions with frozen diffusion augmentation instead of modifying the SSL loss. Second, we replace pure tail selection with a histogram-based mode-window rule that better targets semantically recoverable sparse regions. Third, we provide a broad empirical study across five source regimes, seven downstream tasks, three seeds, competitive long-tailed SSL baselines, and five targeted ablation groups.

2 Related Work

Self-supervised visual representation learning. Modern vision SSL is dominated by contrastive methods such as SimCLR and MoCo [Chen et al., 2020, He et al., 2020], self-distillation methods such as DINO [Caron et al., 2021], and masked prediction methods such as MAE [He et al., 2022]. These approaches have made unlabeled pretraining routine, but they typically hold the source dataset fixed and place the burden of improvement on the learning objective.

Long-tailed and imbalanced SSL. Long-tailed data changes both optimization and representation geometry. TS addresses this through a schedule over the contrastive temperature [Kukleva et al., 2023]. SDCLR prunes a self-competitor to emphasize hard and tail examples [Jiang et al., 2021]. FASSL introduces frequency-aware SSL training [Lin et al., 2023]. COLT injects external OOD data into long-tailed SSL [Bai et al., 2023]. We target a different lever. BRIDGE does not require class frequencies, does not alter the core SSL loss, and does not require a separate external OOD collection. Instead, we repair sparse regions using images generated from the data already being trained on.

OOD scoring and data curation. Data-centric AI emphasizes that dataset composition can matter as much as model design [Whang et al., 2023]. In our setting, the scoring rule must work without labels, density estimation assumptions, or class-frequency estimates. Mean deep nearest-neighbor distance satisfies these constraints and has been effective for OOD detection in high-dimensional representation spaces [Sun et al., 2022]. We use this signal not only to detect sparsity, but to drive targeted repair of underrepresented regions.

Diffusion based augmentation. Diffusion models enable high-quality image synthesis with strong semantic control [Ho et al., 2020, Rombach et al., 2022]. Prior long-tailed work often uses generated images in supervised settings. We instead use a frozen image-to-image diffusion model as a repair operator inside an SSL loop, and we target generation to sparse representation regions rather than to globally sampled prompts.

3 Method

3.1 Setting and repair loop

Let $D^{(0)} = \{x_i\}_{i=1}^{N_0}$ denote an unlabeled source dataset and let f_θ be an SSL encoder trained with objective $\mathcal{L}_{\text{ssl}}$. BRIDGE alternates between pretraining and dataset repair over C cycles. At cycle c, the current encoder is $f_{\theta^{(c)}}$, the current source set is $D^{(c)} = \{x_i\}_{i=1}^{N^{(c)}}$, and the corresponding embedding set is $Z^{(c)} = \{z_i^{(c)}\}_{i=1}^{N^{(c)}}$.

After training on $D^{(c)}$, we embed every image and compute a nonparametric OOD score from mean kNN distance,

$$\begin{aligned} z_i^{(c)} &= f_{\theta^{(c)}}(x_i), \\ d_i^{(c)} &= \frac{1}{k} \sum_{z \in \mathcal{N}_k\left(z_i^{(c)};\, Z^{(c)} \setminus \{z_i^{(c)}\}\right)} \left\| z_i^{(c)} - z \right\|_2^2, \end{aligned} \tag{1}$$

where $\mathcal{N}_k(z; Z)$ denotes the set of the k nearest neighbors of z in Z. Large values of $d_i^{(c)}$ indicate that an image lies in a sparse part of the learned representation space.

3.2 Mode-window OOD selection

The older version of our method selected the largest OOD scores directly. That rule is simple but brittle. The farthest samples are often visually noisy, outlying even relative to the sparse regions that remain semantically useful for repair.

Our current selector therefore works in two steps. First, it clips the empirical OOD distribution to its 1% to 99% quantile range and builds a histogram over the clipped values. Let $b^\star$ denote the modal bin and let $m^{(c)}$ denote the center of that bin. Second, given an augmentation budget B, we select the subset of B indices whose OOD scores are closest to $m^{(c)}$:

$$S^{(c)} = \underset{S \subseteq [N^{(c)}],\, |S| = B}{\arg\min} \sum_{i \in S} \left| d_i^{(c)} - m^{(c)} \right|, \tag{2}$$

where $[N^{(c)}] = \{1, \ldots, N^{(c)}\}$. This yields a symmetric selection around the densest sparse region rather than a direct grab of the farthest tail. Figure 2 shows that this modal sparse region moves across repair cycles, which is why we anchor selection to the current histogram mode instead of repeatedly sampling the farthest tail.

3.3 Diffusion based repair

Given the selected set $S^{(c)}$, we generate G synthetic variants per selected image with a frozen image-to-image diffusion model g_ϕ. In the main method we use Stable Diffusion 3. The generation ablation replaces it with FLUX or with a no-generation re-population baseline. We first form the generated set

$$\begin{aligned} \widetilde{D}^{(c)} &= \left\{ g_\phi(x_i, \epsilon_j) \,\middle|\, i \in S^{(c)},\, j \in [G] \right\}, \\ D^{(c+1)} &= D^{(c)} \cup \widetilde{D}^{(c)}, \end{aligned} \tag{3}$$

where $[G] = \{1, \ldots, G\}$. Because the generator is frozen, the learned component remains the SSL encoder. The repair loop is therefore compatible with any SSL objective that can provide embeddings for OOD scoring.

3.4 Design rationale

The key choice is that we do not try to densify the most isolated samples at all costs. The farthest tail often contains degenerate crops, artifacts, or images whose semantics are too unstable for reliable image-to-image augmentation. The histogram mode targets a more useful region: images that are underrepresented relative to the main body of representation space, yet still belong to a coherent local neighborhood. In the completed selection ablation, this change improves the full seven-task average over the older top-tail rule for both linear-probe and kNN transfer. Uniform sampling remains competitive and even slightly stronger on average in that isolated ablation, so we interpret the mode-window rule narrowly but clearly: it is a principled replacement for extreme-tail selection, not a universally dominant selector.

4 Experimental Setup

Source datasets. We pretrain on five unlabeled source regimes. Three are long-tailed image classification datasets, namely ImageNet-100-LT, CIFAR-10-LT, and CIFAR-100-LT. Two are web-source unlabeled corpora, namely PASS [Asano et al., 2021] and DiffusionDB [Wang et al., 2023]. For PASS and DiffusionDB we use 10k example subsets and override the pretraining split to $[1.0, 0.0, 0.0]$, so all sampled images are used for SSL pretraining.

Downstream evaluation. We evaluate transfer on seven benchmarks: CIFAR-10-LT, CIFAR-100-LT, Stanford Cars [Krause et al., 2013], FGVC Aircraft [Maji et al., 2013], Oxford Flowers [Nilsback and Zisserman, 2008], Oxford-IIIT Pets [Parkhi et al., 2012], and ImageNet-100-LT. In the main text, we report full linear-probe comparisons, together with kNN corroboration for the source-regime comparisons and the combined ablation table. The appendix provides the complete expanded linear-probe and kNN tables for every setting.

Training protocol. All reported numbers are mean $\pm$ standard deviation over three seeds. The main backbone is ResNet-50. Unless noted otherwise, BRIDGE uses $k = 100$ for OOD scoring, selects 500 images per cycle, generates 5 images per selected image, and trains for 5 cycles with 100 epochs per cycle. The cycle ablation changes the number of cycles while adjusting the per-cycle budget accordingly. The successful DINO ablation uses two local crops. Because the downstream tasks are heterogeneous, we emphasize per-dataset tables and use the average only as a compact summary statistic. Appendix A gives additional implementation details, runtime summaries, and asset notes.

Compared methods. We compare SimCLR, TS, SDCLR, BRIDGE, and two hybrids that combine BRIDGE with TS or SDCLR. These are meaningful comparisons because BRIDGE changes the data while TS and SDCLR change the optimization dynamics.

Ablation protocol. All ablations are run on ImageNet-100-LT source pretraining with the same downstream suite and three-seed evaluation. Each ablation changes one component at a time. The pretraining ablation swaps SimCLR, MoCo, and DINO. The generation ablation compares Stable Diffusion 3, FLUX, and no-generation re-population. The selection ablation compares the new mode-window rule, the old top-tail rule that takes the most OOD examples directly, and uniform sampling. The cycle ablation varies the number of repair cycles. The architecture ablation compares ResNet-18, ResNet-50, ViT-S, and ViT-B.

5 Results

5.1 Balanced and imbalanced baselines

Table 1 establishes the basic comparison on ImageNet-100-LT source pretraining. Moving from balanced to imbalanced SimCLR changes the seven-task average only marginally, so the main issue

Table 1: Linear-probe comparison between balanced pretraining, imbalanced pretraining, and BRIDGE on ImageNet-100-LT. The reported average is over all seven downstream linear-probe tasks.

Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Balanced	65.3 ± 1.2	37.0 ± 0.8	8.0 ± 0.5	15.1 ± 0.7	22.4 ± 1.6	**31.2 ± 2.0**	39.0 ± 1.0	31.1 ± 1.1
Imbalanced	65.4 ± 4.2	38.6 ± 1.4	7.6 ± 1.1	15.3 ± 0.9	23.7 ± 1.0	28.4 ± 1.3	39.6 ± 0.3	31.2 ± 1.4
BRIDGE (ours)	**70.7 ± 1.7**	**43.3 ± 0.6**	**9.9 ± 1.5**	**16.4 ± 0.9**	**26.6 ± 2.7**	**31.2 ± 3.7**	**42.0 ± 1.2**	**34.3 ± 1.8**

Table 2: Linear-probe transfer for the three label-derived long-tailed source regimes. The average is computed over all seven downstream linear-probe tasks.

Source	Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ImageNet-100-LT	SimCLR	73.9 ± 0.5	46.8 ± 0.4	13.7 ± 0.3	19.6 ± 0.2	35.9 ± 7.2	38.6 ± 1.5	50.4 ± 0.5	39.8 ± 1.5
	TS	72.5 ± 0.4	44.0 ± 0.5	12.5 ± 1.2	15.0 ± 0.1	35.7 ± 1.8	40.6 ± 2.2	**56.7 ± 1.2**	39.6 ± 1.1
	SDCLR	41.1 ± 1.7	18.0 ± 0.6	1.4 ± 0.3	2.3 ± 0.5	10.8 ± 1.7	10.4 ± 0.3	21.9 ± 0.6	15.1 ± 0.8
	BRIDGE (ours)	**75.7 ± 0.2**	**48.8 ± 0.3**	**16.1 ± 1.0**	**22.4 ± 0.3**	**36.4 ± 2.4**	**42.7 ± 1.9**	53.7 ± 0.6	**42.3 ± 1.0**
	BRIDGE+TS (ours)	73.2 ± 0.1	45.5 ± 0.4	9.5 ± 1.2	13.0 ± 0.7	26.4 ± 2.7	34.4 ± 1.7	47.6 ± 0.6	35.7 ± 1.0
	BRIDGE+SDCLR (ours)	41.5 ± 0.5	17.7 ± 0.4	1.3 ± 0.3	2.4 ± 0.7	12.1 ± 0.8	10.5 ± 1.9	20.7 ± 1.0	15.2 ± 0.8
CIFAR-10-LT	SimCLR	67.8 ± 0.2	38.0 ± 1.0	6.5 ± 0.4	16.0 ± 0.9	19.2 ± 1.6	24.0 ± 1.3	32.0 ± 0.7	29.1 ± 0.9
	TS	68.5 ± 1.3	36.0 ± 1.7	4.3 ± 0.2	10.3 ± 0.1	13.1 ± 2.0	19.7 ± 2.3	31.5 ± 1.2	26.2 ± 1.3
	SDCLR	39.6 ± 1.1	15.6 ± 0.2	1.1 ± 0.1	2.4 ± 0.9	12.0 ± 1.6	9.1 ± 1.8	16.5 ± 1.6	13.8 ± 1.1
	BRIDGE (ours)	70.7 ± 2.1	41.5 ± 0.6	**8.6 ± 0.4**	**17.0 ± 0.9**	**24.0 ± 3.2**	**29.4 ± 2.1**	**34.4 ± 1.5**	**32.2 ± 1.6**
	BRIDGE+TS (ours)	**73.1 ± 1.0**	**42.0 ± 0.3**	4.4 ± 0.8	12.4 ± 0.5	10.2 ± 1.6	23.2 ± 1.2	34.2 ± 1.1	28.5 ± 0.9
	BRIDGE+SDCLR (ours)	39.1 ± 0.2	15.3 ± 1.0	1.1 ± 0.3	2.1 ± 0.6	8.9 ± 2.6	9.5 ± 1.1	16.8 ± 0.7	13.3 ± 0.9
CIFAR-100-LT	SimCLR	59.0 ± 1.5	31.1 ± 1.6	4.1 ± 0.3	11.2 ± 1.0	12.6 ± 3.4	20.1 ± 1.0	28.7 ± 1.3	23.8 ± 1.4
	TS	56.3 ± 0.2	28.0 ± 0.1	2.8 ± 0.6	8.5 ± 0.2	15.5 ± 2.5	16.2 ± 3.1	26.2 ± 1.8	21.9 ± 1.2
	SDCLR	37.1 ± 0.8	15.7 ± 0.2	1.9 ± 0.1	2.4 ± 0.7	9.4 ± 1.0	9.9 ± 1.7	16.2 ± 0.3	13.2 ± 0.7
	BRIDGE (ours)	**66.8 ± 3.1**	**38.3 ± 2.2**	**6.2 ± 1.8**	**14.6 ± 1.6**	**19.8 ± 0.4**	**22.7 ± 2.3**	**32.6 ± 1.4**	**28.7 ± 1.8**
	BRIDGE+TS (ours)	66.4 ± 3.3	38.0 ± 3.7	4.2 ± 1.0	10.9 ± 0.7	17.2 ± 3.6	21.6 ± 3.3	31.4 ± 2.3	27.1 ± 2.6
	BRIDGE+SDCLR (ours)	39.1 ± 1.9	14.9 ± 0.9	0.9 ± 0.2	3.0 ± 0.4	10.9 ± 2.5	7.9 ± 0.7	16.0 ± 0.3	13.2 ± 1.0

is not solved by simply removing majority examples. By contrast, BRIDGE improves the seven-task average linear-probe score by 3.05 points over imbalanced SimCLR and by 3.14 points over balanced SimCLR. The gains are spread across the suite rather than concentrated on a single target. Relative to imbalanced SimCLR, BRIDGE improves CIFAR-10-LT by 5.37 points, CIFAR-100-LT by 4.71 points, Cars by 2.30 points, Flowers by 2.83 points, and ImageNet-100-LT transfer by 2.35 points, while matching the best Pets score. This is the first indication that repairing sparse regions of representation space is more effective than changing the class balance of the source set alone.

5.2 Comparison across source datasets

Table 2 reports the three label-derived long-tailed source regimes with all seven downstream tasks shown directly. BRIDGE is the strongest method on the overall average in all three regimes. The clearest case is CIFAR-100-LT source pretraining, where BRIDGE improves the seven-task average over source-matched SimCLR by 4.89 points and is best on every downstream dataset. On ImageNet-100-LT, TS remains strongest on the in-domain ImageNet-100-LT transfer column, but BRIDGE wins on the remaining six targets and on the overall average. On CIFAR-10-LT, BRIDGE+TS slightly improves the two CIFAR targets, while BRIDGE remains strongest on the fine-grained tasks and on the overall average. Across the three regimes, the same pattern emerges: optimization changes can help on a source-matched target, but repairing the source set produces the strongest broad transfer.

Table 3 shows that this ranking is not specific to linear probing. Under kNN transfer, BRIDGE remains the strongest average method in all three label-derived regimes. On ImageNet-100-LT it is best on six of the seven downstream tasks, with TS retaining only the in-domain ImageNet-100-LT column. On CIFAR-10-LT and CIFAR-100-LT, BRIDGE is best on every reported kNN target. This matters because kNN removes the learned linear head from the evaluation. The gains therefore reflect changes in feature geometry itself, not only easier linear readout.

Table 4 evaluates the two web-source regimes, which are qualitatively different from the label-derived long-tailed datasets. PASS is a natural web image corpus, while DiffusionDB is fully synthetic. BRIDGE is again the strongest average method in both settings. On PASS-10k, it improves the seven-task average over source-matched SimCLR by 3.15 points and is best on five of the seven downstream tasks. On DiffusionDB-10k, it improves the average by 2.54 points and is best

Table 3: kNN transfer for the three label-derived long-tailed source regimes. The average is computed over all seven downstream kNN tasks.

Source	Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ImageNet-100-LT	SimCLR	68.6 ± 0.4	37.3 ± 0.2	6.0 ± 0.7	11.2 ± 0.4	32.2 ± 0.7	23.6 ± 0.3	43.9 ± 0.9	31.8 ± 0.5
	TS	68.4 ± 0.3	36.5 ± 0.1	4.7 ± 0.2	8.3 ± 0.5	26.6 ± 1.0	23.6 ± 1.7	$\mathbf{52.2 \pm 0.5}$	31.5 ± 0.6
	SDCLR	32.5 ± 0.4	10.9 ± 0.2	1.0 ± 0.3	2.1 ± 0.1	10.7 ± 0.7	4.7 ± 0.2	11.2 ± 0.2	10.4 ± 0.3
	BRIDGE (ours)	$\mathbf{68.9 \pm 0.5}$	$\mathbf{37.8 \pm 0.1}$	$\mathbf{6.9 \pm 0.1}$	$\mathbf{12.1 \pm 0.2}$	$\mathbf{36.0 \pm 0.1}$	$\mathbf{25.3 \pm 0.9}$	48.2 ± 0.4	$\mathbf{33.6 \pm 0.3}$
	BRIDGE+TS (ours)	65.2 ± 0.2	33.7 ± 0.5	2.9 ± 0.4	6.6 ± 0.2	18.6 ± 0.4	15.8 ± 1.0	40.5 ± 0.8	26.2 ± 0.5
	BRIDGE+SDCLR (ours)	33.8 ± 0.4	11.5 ± 0.5	1.3 ± 0.4	2.0 ± 0.1	10.2 ± 0.7	4.9 ± 0.2	12.6 ± 0.6	10.9 ± 0.4
CIFAR-10-LT	SimCLR	66.9 ± 0.4	33.4 ± 0.3	3.7 ± 0.4	9.8 ± 0.7	16.5 ± 0.8	13.9 ± 0.8	24.0 ± 0.8	24.0 ± 0.6
	TS	66.3 ± 0.6	27.8 ± 0.4	2.4 ± 0.6	6.4 ± 0.2	10.5 ± 0.7	11.1 ± 0.4	22.0 ± 0.9	20.9 ± 0.5
	SDCLR	30.1 ± 0.3	10.0 ± 0.3	1.7 ± 0.3	2.2 ± 0.2	10.1 ± 0.7	4.4 ± 0.4	9.8 ± 0.3	9.7 ± 0.4
	BRIDGE (ours)	$\mathbf{69.0 \pm 1.2}$	$\mathbf{35.2 \pm 0.9}$	$\mathbf{4.2 \pm 0.6}$	$\mathbf{10.4 \pm 0.5}$	$\mathbf{17.0 \pm 0.6}$	$\mathbf{15.2 \pm 0.8}$	$\mathbf{25.2 \pm 0.5}$	$\mathbf{25.2 \pm 0.7}$
	BRIDGE+TS (ours)	67.7 ± 0.8	28.0 ± 0.8	2.1 ± 0.2	5.1 ± 0.7	8.1 ± 1.0	9.1 ± 0.8	21.8 ± 0.5	20.3 ± 0.7
	BRIDGE+SDCLR (ours)	31.3 ± 1.2	10.4 ± 0.5	1.0 ± 0.1	1.8 ± 0.3	9.6 ± 1.3	4.5 ± 0.1	10.1 ± 0.4	9.8 ± 0.6
CIFAR-100-LT	SimCLR	59.3 ± 0.8	30.1 ± 0.9	3.0 ± 0.3	7.4 ± 0.4	13.7 ± 0.2	10.8 ± 0.4	21.8 ± 1.1	20.9 ± 0.6
	TS	53.1 ± 0.6	24.6 ± 0.4	2.4 ± 0.4	5.5 ± 0.3	9.3 ± 0.4	8.3 ± 0.2	18.7 ± 1.0	17.4 ± 0.5
	SDCLR	25.7 ± 1.2	8.5 ± 0.6	0.9 ± 0.4	1.9 ± 0.2	7.3 ± 0.3	3.9 ± 0.3	8.3 ± 1.1	8.1 ± 0.6
	BRIDGE (ours)	$\mathbf{62.5 \pm 0.8}$	$\mathbf{32.5 \pm 0.3}$	$\mathbf{3.7 \pm 0.3}$	$\mathbf{8.7 \pm 0.5}$	$\mathbf{14.1 \pm 0.9}$	$\mathbf{12.1 \pm 0.7}$	$\mathbf{23.0 \pm 1.2}$	$\mathbf{22.4 \pm 0.7}$
	BRIDGE+TS (ours)	56.3 ± 1.2	27.8 ± 0.5	1.9 ± 0.0	4.8 ± 0.6	8.0 ± 1.0	9.0 ± 0.6	20.8 ± 0.8	18.4 ± 0.7
	BRIDGE+SDCLR (ours)	29.7 ± 2.2	9.8 ± 1.5	1.4 ± 0.4	2.3 ± 0.2	9.2 ± 1.1	4.8 ± 0.1	8.9 ± 0.6	9.4 ± 0.9

Table 4: Linear-probe transfer for the two web-source regimes. PASS is a natural web-image corpus, while DiffusionDB is fully synthetic. The average is computed over all seven downstream linear-probe tasks.

Source	Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
PASS-10k	SimCLR	72.3 ± 0.2	45.5 ± 0.5	11.0 ± 0.2	17.5 ± 1.4	33.3 ± 1.3	33.3 ± 0.9	44.7 ± 0.9	36.8 ± 0.8
	TS	71.5 ± 0.2	43.5 ± 0.4	10.2 ± 1.2	17.2 ± 0.2	33.1 ± 2.1	$\mathbf{36.4 \pm 1.8}$	49.4 ± 0.4	37.3 ± 0.9
	SDCLR	42.2 ± 1.1	17.7 ± 0.8	1.8 ± 0.3	2.2 ± 0.8	10.6 ± 1.3	8.2 ± 1.6	20.8 ± 0.4	14.8 ± 0.9
	BRIDGE (ours)	$\mathbf{74.2 \pm 0.2}$	$\mathbf{47.4 \pm 0.7}$	$\mathbf{13.0 \pm 1.2}$	$\mathbf{20.4 \pm 0.4}$	$\mathbf{39.4 \pm 3.4}$	35.8 ± 0.1	49.5 ± 1.0	$\mathbf{40.0 \pm 1.0}$
	BRIDGE+TS (ours)	69.1 ± 0.4	40.5 ± 0.8	12.1 ± 0.5	14.3 ± 0.3	32.7 ± 4.1	30.8 ± 1.0	$\mathbf{52.6 \pm 0.5}$	36.0 ± 1.1
	BRIDGE+SDCLR (ours)	45.5 ± 1.0	19.3 ± 0.2	1.5 ± 0.7	3.3 ± 0.2	11.0 ± 1.6	11.3 ± 1.2	20.9 ± 0.9	16.1 ± 0.8
DiffusionDB-10k	SimCLR	74.4 ± 0.5	48.6 ± 0.2	10.9 ± 0.7	17.6 ± 0.8	31.2 ± 2.4	36.1 ± 1.3	45.3 ± 0.5	37.7 ± 0.9
	TS	72.3 ± 0.3	44.4 ± 0.3	10.7 ± 0.6	15.4 ± 0.6	33.1 ± 1.1	$\mathbf{39.9 \pm 2.7}$	48.8 ± 0.2	37.8 ± 0.8
	SDCLR	42.4 ± 0.2	18.4 ± 0.2	1.1 ± 0.2	2.4 ± 0.1	13.0 ± 3.7	8.2 ± 1.4	21.0 ± 0.6	15.2 ± 0.9
	BRIDGE (ours)	$\mathbf{76.2 \pm 0.1}$	$\mathbf{49.8 \pm 0.1}$	$\mathbf{12.4 \pm 1.0}$	$\mathbf{20.4 \pm 0.2}$	$\mathbf{36.1 \pm 2.7}$	38.0 ± 3.2	48.9 ± 0.8	$\mathbf{40.3 \pm 1.2}$
	BRIDGE+TS (ours)	68.8 ± 0.1	40.2 ± 0.4	9.2 ± 1.1	12.8 ± 0.5	31.2 ± 2.4	34.8 ± 1.6	$\mathbf{51.0 \pm 0.6}$	35.4 ± 1.0
	BRIDGE+SDCLR (ours)	42.2 ± 0.7	17.6 ± 0.3	1.1 ± 0.6	2.4 ± 0.1	11.9 ± 3.6	11.7 ± 1.3	21.3 ± 0.8	15.5 ± 1.1

on CIFAR-10-LT, CIFAR-100-LT, Cars, Aircraft, and Flowers. The DiffusionDB result is especially informative. Because the source data are already synthetic, the gains cannot be attributed only to harvesting more natural web diversity. They instead indicate that the repair loop remains useful even when the source corpus itself carries strong generative and stylistic bias.

The same conclusion holds under nonparametric evaluation. Table 5 reports full kNN transfer for the two web-source regimes. On PASS-10k, BRIDGE improves the seven-task kNN average by 1.32 points over SimCLR and is best on six of the seven downstream tasks, with SimCLR slightly better only on Cars. On DiffusionDB-10k, BRIDGE is best on all seven downstream tasks and improves the kNN average by 1.58 points. Together with Table 4, this shows that the effect is not tied to a learned probe and not restricted to natural image corpora.

5.3 Ablations

We now examine each design choice separately on ImageNet-100-LT source pretraining. Each ablation changes one component of BRIDGE while holding the rest of the pipeline fixed. Table 6 consolidates all five ablation groups in one full-width table so that every downstream dataset remains visible in the main text. The appendix provides the same groups again as separate linear-probe and kNN tables for readers who want the expanded view. The purpose of this section is not only to show that BRIDGE works, but to identify which parts of the loop are actually responsible for the gain.

SSL objective. SimCLR is the strongest objective inside the BRIDGE loop on every downstream dataset and on the overall average. MoCo is consistently weaker, and DINO remains the weakest option even after the final configuration that reduces the number of local crops from six to two. In this setting, the repair loop works best with a standard contrastive objective rather than momentum

Table 5: kNN transfer for the two web-source regimes. PASS is a natural web-image corpus, while DiffusionDB is fully synthetic. The average is computed over all seven downstream kNN tasks.

Source	Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
PASS-10k	SimCLR	67.2 ± 0.2	37.0 ± 0.3	$\mathbf{5.4 \pm 0.3}$	10.3 ± 0.5	30.4 ± 0.0	16.7 ± 0.5	38.3 ± 0.3	29.3 ± 0.3
	TS	65.2 ± 0.7	34.6 ± 0.3	4.4 ± 0.2	7.9 ± 0.2	24.5 ± 0.1	14.4 ± 0.4	39.3 ± 0.4	27.2 ± 0.3
	SDCLR	32.4 ± 1.5	11.5 ± 0.8	1.3 ± 0.2	2.0 ± 0.1	8.6 ± 0.7	5.2 ± 0.2	11.1 ± 0.8	10.3 ± 0.6
	BRIDGE (ours)	$\mathbf{67.8 \pm 0.6}$	$\mathbf{37.1 \pm 0.5}$	5.2 ± 0.4	$\mathbf{10.8 \pm 0.3}$	$\mathbf{33.7 \pm 0.6}$	$\mathbf{17.9 \pm 0.3}$	$\mathbf{41.9 \pm 0.4}$	$\mathbf{30.6 \pm 0.5}$
	BRIDGE+TS (ours)	59.0 ± 0.4	27.8 ± 0.5	3.3 ± 0.4	6.6 ± 0.0	23.3 ± 0.4	13.9 ± 0.3	39.2 ± 0.3	24.7 ± 0.3
	BRIDGE+SDCLR (ours)	37.9 ± 0.7	13.6 ± 0.0	1.2 ± 0.2	2.7 ± 0.1	8.9 ± 1.3	6.3 ± 0.1	13.9 ± 1.1	12.1 ± 0.5
DiffusionDB-10k	SimCLR	67.3 ± 0.1	36.2 ± 0.4	4.3 ± 0.2	8.8 ± 0.1	26.8 ± 0.4	17.3 ± 0.1	36.7 ± 0.8	28.2 ± 0.3
	TS	64.7 ± 0.4	33.6 ± 0.3	3.1 ± 0.5	7.3 ± 0.3	23.2 ± 0.4	17.0 ± 0.0	37.8 ± 0.6	26.7 ± 0.4
	SDCLR	32.6 ± 1.2	10.0 ± 0.3	1.3 ± 0.1	2.2 ± 0.2	6.1 ± 0.7	4.8 ± 0.4	10.6 ± 0.5	9.7 ± 0.5
	BRIDGE (ours)	$\mathbf{68.1 \pm 0.3}$	$\mathbf{37.0 \pm 0.2}$	$\mathbf{5.4 \pm 0.5}$	$\mathbf{9.5 \pm 0.1}$	$\mathbf{30.4 \pm 0.4}$	$\mathbf{18.5 \pm 0.1}$	$\mathbf{39.9 \pm 0.6}$	$\mathbf{29.8 \pm 0.3}$
	BRIDGE+TS (ours)	57.2 ± 0.7	25.6 ± 1.0	2.8 ± 0.3	6.8 ± 0.1	22.3 ± 0.5	15.4 ± 0.6	37.3 ± 0.6	23.9 ± 0.6
	BRIDGE+SDCLR (ours)	37.1 ± 1.3	11.8 ± 0.9	1.3 ± 0.1	2.2 ± 0.2	6.8 ± 0.5	5.9 ± 0.7	12.5 ± 0.7	11.1 ± 0.6

Table 6: Main ablations on ImageNet-100-LT. Each block changes one component while holding the remaining pipeline fixed. The average is computed over all seven downstream linear-probe tasks.

Group	Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
SSL objective	SimCLR	$\mathbf{71.8 \pm 1.2}$	$\mathbf{43.5 \pm 0.5}$	$\mathbf{9.0 \pm 0.5}$	$\mathbf{16.9 \pm 0.3}$	$\mathbf{27.2 \pm 1.5}$	$\mathbf{33.2 \pm 3.3}$	$\mathbf{41.7 \pm 1.3}$	$\mathbf{34.7 \pm 1.2}$
	MoCo	46.6 ± 3.1	18.4 ± 2.0	3.1 ± 0.5	4.3 ± 0.8	10.4 ± 3.2	15.0 ± 1.8	26.6 ± 1.9	17.8 ± 1.9
	DINO	43.6 ± 0.7	17.8 ± 0.6	1.3 ± 0.5	1.6 ± 0.2	12.6 ± 4.4	8.7 ± 1.6	20.6 ± 1.1	15.2 ± 1.3
Generation	Stable Diffusion 3	$\mathbf{75.0 \pm 0.2}$	$\mathbf{48.4 \pm 0.2}$	$\mathbf{15.1 \pm 1.2}$	$\mathbf{20.6 \pm 0.7}$	$\mathbf{37.4 \pm 2.0}$	$\mathbf{41.2 \pm 0.7}$	$\mathbf{51.8 \pm 1.1}$	$\mathbf{41.4 \pm 0.8}$
	FLUX	70.7 ± 0.4	42.1 ± 1.4	10.3 ± 1.3	17.5 ± 0.9	29.4 ± 4.6	31.5 ± 2.9	43.7 ± 1.4	35.0 ± 1.9
	No-generation re-population	67.0 ± 2.6	39.4 ± 2.2	8.0 ± 0.6	15.4 ± 0.5	25.7 ± 1.4	31.7 ± 3.8	40.6 ± 0.5	32.6 ± 1.7
Selection	Mode-window	$\mathbf{71.9 \pm 0.3}$	43.4 ± 0.7	9.0 ± 0.5	16.9 ± 0.2	27.9 ± 3.1	31.6 ± 3.3	41.7 ± 1.3	34.6 ± 1.3
	Top-tail	70.8 ± 0.5	43.1 ± 1.0	$\mathbf{10.0 \pm 1.7}$	15.9 ± 1.3	27.7 ± 1.0	30.5 ± 1.2	$\mathbf{42.2 \pm 1.4}$	34.3 ± 1.2
	Uniform	71.5 ± 0.2	$\mathbf{43.8 \pm 0.7}$	9.3 ± 0.9	$\mathbf{17.5 \pm 1.2}$	$\mathbf{32.9 \pm 0.4}$	$\mathbf{31.7 \pm 3.9}$	42.0 ± 0.9	$\mathbf{35.5 \pm 1.2}$
Cycles	2 cycles	67.9 ± 0.6	40.7 ± 0.6	9.4 ± 1.8	16.0 ± 1.5	$\mathbf{30.5 \pm 3.2}$	27.4 ± 1.7	40.1 ± 0.4	33.1 ± 1.4
	5 cycles	70.7 ± 1.7	43.3 ± 0.6	$\mathbf{9.9 \pm 1.5}$	16.4 ± 0.9	26.6 ± 2.7	31.2 ± 3.7	42.0 ± 1.2	34.3 ± 1.8
	10 cycles	70.5 ± 1.2	43.8 ± 0.6	9.2 ± 0.9	17.2 ± 1.6	24.8 ± 5.2	31.3 ± 1.7	41.9 ± 0.6	34.1 ± 1.7
	20 cycles	$\mathbf{71.6 \pm 1.1}$	$\mathbf{44.6 \pm 0.9}$	9.8 ± 0.3	$\mathbf{17.3 \pm 1.2}$	27.0 ± 2.1	$\mathbf{33.3 \pm 4.4}$	$\mathbf{42.1 \pm 1.5}$	$\mathbf{35.1 \pm 1.6}$
Architecture	ResNet-18	67.7 ± 0.8	40.4 ± 0.5	$\mathbf{10.6 \pm 0.7}$	15.4 ± 0.3	26.1 ± 2.4	29.5 ± 2.7	41.0 ± 1.2	32.9 ± 1.2
	ResNet-50	$\mathbf{71.7 \pm 1.0}$	$\mathbf{44.0 \pm 1.1}$	9.0 ± 0.5	$\mathbf{17.4 \pm 0.7}$	28.8 ± 1.7	$\mathbf{32.0 \pm 1.6}$	41.7 ± 1.3	$\mathbf{34.9 \pm 1.1}$
	ViT-S	62.7 ± 0.4	37.6 ± 0.6	8.8 ± 0.5	12.2 ± 0.2	35.1 ± 4.2	30.2 ± 1.6	42.5 ± 1.2	32.7 ± 1.3
	ViT-B	66.0 ± 2.3	40.6 ± 2.3	8.3 ± 0.5	14.3 ± 0.8	$\mathbf{38.8 \pm 2.6}$	31.3 ± 2.1	$\mathbf{44.5 \pm 1.7}$	34.8 ± 1.8

contrast or self-distillation. This is useful practically because it shows that the gains do not rely on an unusually specialized SSL backbone.

Generation mechanism. Stable Diffusion 3 is best on every downstream dataset and improves the seven-task average by 8.81 points over re-population. FLUX is better than simple re-population, but it remains well below Stable Diffusion 3 on both the fine-grained tasks and the in-domain target. This ablation shows that BRIDGE needs semantically faithful image-to-image generation, not only more images or a generic synthetic source expansion.

Sample selection. The selection block answers the main design question raised by the earlier version of our method. We do not compare the new mode-window rule only to uniform sampling. We also compare it directly to the original pure-tail strategy that takes the most OOD examples. On that comparison, the mode-window selector is better on the overall average and on CIFAR-10-LT, Aircraft, and Pets, while the top-tail rule retains only Cars and ImageNet-100-LT. Uniform sampling remains competitive and is slightly stronger on the overall average in this isolated ablation. The right interpretation is therefore specific: the histogram-based selector is a better replacement for extreme-tail selection, even if it is not universally better than every alternative selector.

Cycle count. Performance improves as the repair loop is allowed to iterate longer. Two cycles already recover some benefit, but 20 cycles produce the strongest average transfer and the best results on CIFAR-10-LT, CIFAR-100-LT, Aircraft, Pets, and ImageNet-100-LT. We nevertheless keep 5 cycles as the default in the broad source-regime comparison because it is the smallest setting that remains consistently strong while staying tractable across all source regimes.

Encoder architecture. ResNet-50 gives the best overall average and is strongest on CIFAR-10-LT, CIFAR-100-LT, Aircraft, and Pets. ViT-B is best on Flowers and ImageNet-100-LT, while ResNet-

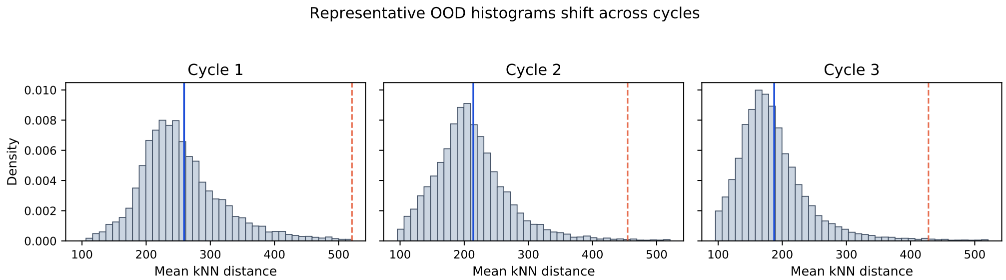

Figure 2: Representative OOD histograms across BRIDGE cycles from a successful ImageNet-100-LT run. The modal shoulder of the distribution moves as training and repair progress, while the extreme tail remains sparse. This supports selecting around the modal sparse band rather than repeatedly taking the farthest tail samples.

Table 7: Main kNN ablations on ImageNet-100-LT. Each block changes one component while holding the remaining pipeline fixed. The average is computed over all seven downstream kNN tasks.

Group	Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
SSL objective	SimCLR	**66.3 ± 0.8**	**35.3 ± 0.4**	**5.1 ± 0.7**	**9.5 ± 0.1**	**24.8 ± 0.6**	**18.9 ± 1.0**	**36.7 ± 0.7**	**28.1 ± 0.6**
	MoCo	56.4 ± 1.6	25.9 ± 0.8	3.2 ± 0.4	5.2 ± 0.4	16.3 ± 1.1	12.2 ± 0.7	24.3 ± 0.7	20.5 ± 0.8
	DINO	34.4 ± 0.2	12.4 ± 0.1	1.6 ± 0.1	2.4 ± 0.1	6.2 ± 0.8	5.4 ± 0.4	10.7 ± 0.8	10.4 ± 0.4
Generation	Stable Diffusion 3	**68.3 ± 0.4**	**36.8 ± 0.2**	**6.5 ± 0.4**	**11.7 ± 0.6**	**35.1 ± 0.4**	**24.2 ± 0.3**	**45.9 ± 0.5**	**32.7 ± 0.4**
	FLUX	65.8 ± 1.0	35.4 ± 0.8	5.7 ± 0.7	9.4 ± 0.9	27.7 ± 1.0	18.8 ± 1.3	38.7 ± 1.2	28.8 ± 1.0
	No-generation re-population	64.5 ± 0.1	33.4 ± 0.6	4.2 ± 0.6	9.0 ± 0.4	23.0 ± 0.5	17.0 ± 0.4	34.2 ± 1.2	26.5 ± 0.5
Selection	Mode-window	**66.3 ± 0.8**	35.3 ± 0.4	**5.1 ± 0.7**	9.5 ± 0.1	24.8 ± 0.6	18.9 ± 1.0	36.7 ± 0.7	28.1 ± 0.6
	Top-tail	65.7 ± 0.8	34.4 ± 0.2	4.1 ± 0.4	9.6 ± 0.2	23.9 ± 0.9	18.1 ± 0.8	35.4 ± 1.2	27.3 ± 0.6
	Uniform	66.1 ± 1.0	**35.7 ± 0.3**	4.9 ± 0.1	**10.2 ± 0.7**	**25.3 ± 0.4**	**19.0 ± 0.8**	**37.4 ± 0.6**	**28.4 ± 0.6**
Cycles	2 cycles	65.1 ± 0.1	34.1 ± 0.5	4.6 ± 0.3	9.3 ± 0.3	23.8 ± 0.3	17.3 ± 0.7	34.5 ± 0.6	26.9 ± 0.4
	5 cycles	66.0 ± 0.9	35.0 ± 0.9	**5.0 ± 0.4**	9.8 ± 0.2	24.8 ± 0.4	18.0 ± 0.5	35.6 ± 1.2	27.7 ± 0.6
	10 cycles	66.4 ± 0.5	35.4 ± 0.3	4.8 ± 0.3	**10.4 ± 0.6**	**25.2 ± 0.8**	**18.6 ± 0.9**	36.0 ± 0.7	**28.1 ± 0.6**
	20 cycles	**66.5 ± 0.7**	**35.6 ± 0.8**	4.8 ± 0.3	9.5 ± 0.4	24.9 ± 0.5	18.6 ± 1.2	**36.3 ± 1.2**	28.0 ± 0.7
Architecture	ResNet-18	65.0 ± 0.3	34.4 ± 0.3	4.7 ± 0.3	**9.6 ± 0.4**	23.5 ± 1.4	16.6 ± 0.3	36.3 ± 1.1	27.2 ± 0.6
	ResNet-50	66.3 ± 0.8	35.3 ± 0.4	5.1 ± 0.7	9.5 ± 0.1	24.8 ± 0.6	18.9 ± 1.0	36.7 ± 0.7	28.1 ± 0.6
	ViT-S	67.3 ± 0.4	37.5 ± 0.3	5.3 ± 0.2	7.6 ± 0.1	36.2 ± 0.7	20.1 ± 1.0	42.1 ± 1.0	30.9 ± 0.5
	ViT-B	**69.3 ± 0.8**	**40.1 ± 0.8**	**6.0 ± 0.2**	8.3 ± 0.2	**39.8 ± 0.8**	**21.9 ± 1.4**	**45.1 ± 0.7**	**32.9 ± 0.7**

18 is slightly best on Cars. The backbone choice therefore shifts the balance between target types, but ResNet-50 remains the most reliable default across the full transfer suite.

kNN corroboration. Table 7 shows the same five ablation blocks under kNN transfer. The main conclusions are stable. SimCLR remains the strongest SSL objective, Stable Diffusion 3 remains the best generator, and the mode-window selector remains better than the original top-tail rule on the overall average. The main difference appears in the backbone block: under kNN, ViT-B becomes the strongest architecture overall, while ResNet-50 remains the best linear-probe default in Table 6. This split is informative. It suggests that ViT-B preserves stronger local neighborhoods, whereas ResNet-50 yields the most reliable linear readout under the completed sweep.

6 Limitations

The reported PASS and DiffusionDB results use aligned 10k source subsets drawn from larger web-source corpora, so they evaluate BRIDGE under a fixed source budget rather than at the full corpus scale. The main comparison also centers on a single default operating point based on ResNet-50, five repair cycles, and one generation budget, while broader architecture and cycle effects appear only in the ablations. Finally, we evaluate representation transfer through linear probing and kNN classification. These are appropriate diagnostics for the data-repair hypothesis, but they do not exhaust the downstream settings in which repaired source data could matter.

7 Conclusion

We find that long-tailed SSL is not only an optimization problem. It is also a data-geometry problem. By repeatedly identifying sparse representation regions and repairing them with frozen diffusion augmentation, BRIDGE improves average transfer across conventional long-tailed sources and across two web-source regimes. The new mode-window selector matters because it replaces an overly literal extreme-tail rule with a more stable target region. Within the completed experimental protocol, BRIDGE is the strongest average transfer method across all five source regimes that we study.

References

Yuki Asano, Christian Rupprecht, Andrew Zisserman, and Andrea Vedaldi. PASS: An ImageNet replacement for self-supervised pretraining without humans. In Joaquin Vanschoren and Serena Yeung, editors, *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, volume 1, 2021. URL https://datasets-benchmarks-proceedings.neurips.cc/paper_files/paper/2021/file/0777d5c17d4066b82ab86dff8a46af6f-Paper-round1.pdf.

Jianhong Bai, Zuozhu Liu, Hualiang Wang, Jin Hao, Yan Feng, Huanpeng Chu, and Haoji Hu. On the effectiveness of out-of-distribution data in self-supervised long-tail learning. In *The Eleventh International Conference on Learning Representations*, 2023. URL https://openreview.net/forum?id=v8JIQdiN9Sh. Published as a conference paper at ICLR 2023.

Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 9650–9660. IEEE, October 2021. doi: 10.1109/ICCV48922.2021.00951. URL https://openaccess.thecvf.com/content/ICCV2021/html/Caron_Emerging_Properties_in_Self-Supervised_Vision_Transformers_ICCV_2021_paper.html.

Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. In Hal Daumé III and Aarti Singh, editors, *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 1597–1607. PMLR, 13–18 Jul 2020. URL https://proceedings.mlr.press/v119/chen20j.html.

Kaiming He, Haoqi Fan, Yuxin Wu, Saining Xie, and Ross Girshick. Momentum contrast for unsupervised visual representation learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 9729–9738. IEEE, June 2020. doi: 10.1109/CVPR42600.2020.00975. URL https://openaccess.thecvf.com/content_CVPR_2020/html/He_Momentum_Contrast_for_Unsupervised_Visual_Representation_Learning_CVPR_2020_paper.html.

Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 16000–16009. IEEE, June 2022. URL https://openaccess.thecvf.com/content/CVPR2022/html/He_Masked_Autoencoders_Are_Scalable_Vision_Learners_CVPR_2022_paper.html.

Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In Hugo Larochelle, Marc'Aurelio Ranzato, Raia Hadsell, Maria-Florina Balcan, and Hsuan-Tien Lin, editors, *Advances in Neural Information Processing Systems*, volume 33, pages 6840–6851. Curran Associates, Inc., 2020. URL https://proceedings.neurips.cc/paper_files/paper/2020/file/4c5bcfec8584af0d967f1ab10179ca4b-Paper.pdf.

Ziyu Jiang, Tianlong Chen, Bobak J. Mortazavi, and Zhangyang Wang. Self-damaging contrastive learning. In Marina Meila and Tong Zhang, editors, *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 4927–4939. PMLR, 18–24 Jul 2021. URL https://proceedings.mlr.press/v139/jiang21a.html.

Jonathan Krause, Michael Stark, Jia Deng, and Li Fei-Fei. 3d object representations for fine-grained categorization. In *2013 IEEE International Conference on Computer Vision Workshops*, pages 554–561. IEEE, 2013. doi: 10.1109/ICCVW.2013.77. URL https://doi.org/10.1109/ICCVW.2013.77.

Anna Kukleva, Moritz Böhle, Bernt Schiele, Hilde Kuehne, and Christian Rupprecht. Temperature schedules for self-supervised contrastive methods on long-tail data. In *The Eleventh International Conference on Learning Representations*, 2023. URL https://openreview.net/forum?id=ejHUr4nfHhD. Published as a conference paper at ICLR 2023.

Ci-Siang Lin, Min-Hung Chen, and Yu-Chiang Frank Wang. Frequency-aware self-supervised long-tailed learning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops*, pages 963–972. IEEE, October 2023. doi: 10.1109/ICCVW60793.2023.00103. URL https://openaccess.thecvf.com/content/ICCV2023W/LIMIT/html/Lin_Frequency-Aware_Self-Supervised_Long-Tailed_Learning_ICCVW_2023_paper.html.

Subhransu Maji, Esa Rahtu, Juho Kannala, Matthew B. Blaschko, and Andrea Vedaldi. Fine-grained visual classification of aircraft, 2013. URL https://arxiv.org/abs/1306.5151.

Maria-Elena Nilsback and Andrew Zisserman. Automated flower classification over a large number of classes. In *2008 Sixth Indian Conference on Computer Vision, Graphics & Image Processing*, pages 722–729. IEEE, 2008. doi: 10.1109/ICVGIP.2008.47. URL https://doi.org/10.1109/ICVGIP.2008.47.

Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, Mahmoud Assran, Nicolas Ballas, Wojciech Galuba, Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat, Vasu Sharma, Gabriel Synnaeve, Hu Xu, Hervé Jégou, Julien Mairal, Patrick Labatut, Armand Joulin, and Piotr Bojanowski. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024. URL https://openreview.net/forum?id=a68SUt6zFt. Published 05/2024.

Omkar M. Parkhi, Andrea Vedaldi, Andrew Zisserman, and C. V. Jawahar. Cats and dogs. In *2012 IEEE Conference on Computer Vision and Pattern Recognition*, pages 3498–3505. IEEE, 2012. doi: 10.1109/CVPR.2012.6248092. URL https://doi.org/10.1109/CVPR.2012.6248092.

Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 10684–10695. IEEE, June 2022. URL https://openaccess.thecvf.com/content/CVPR2022/html/Rombach_High-Resolution_Image_Synthesis_With_Latent_Diffusion_Models_CVPR_2022_paper.html.

Yiyou Sun, Yifei Ming, Xiaojin Zhu, and Yixuan Li. Out-of-distribution detection with deep nearest neighbors. In Kamalika Chaudhuri, Stefanie Jegelka, Le Song, Csaba Szepesvári, Gang Niu, and Sivan Sabato, editors, *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 20827–20840. PMLR, 17–23 Jul 2022. URL https://proceedings.mlr.press/v162/sun22d.html.

Zijie J. Wang, Evan Montoya, David Munechika, Haoyang Yang, Benjamin Hoover, and Duen Horng Chau. DiffusionDB: A large-scale prompt gallery dataset for text-to-image generative models. In Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki, editors, *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 893–911, Toronto, Canada, July 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.acl-long.51. URL https://aclanthology.org/2023.acl-long.51/.

Steven Euijong Whang, Yuji Roh, Hwanjun Song, and Jae-Gil Lee. Data collection and quality challenges in deep learning: A data-centric AI perspective. *The VLDB Journal*, 32(4): 791–813, 2023. doi: 10.1007/s00778-022-00775-9. URL https://doi.org/10.1007/s00778-022-00775-9.

A Implementation and Reproducibility

Core training configuration. The completed main runs use ResNet-50 encoders. Most baseline, BRIDGE, and source-regime comparisons use 5 cycles with 100 epochs per cycle. The default BRIDGE configuration uses mean kNN distance with $k = 100$, selects 500 samples per cycle, and generates 5 images per selected sample. The default optimizer is AdamW with weight decay 10^{-4}, and method-specific learning rates and temperatures follow the completed job configurations.

OOD scoring and selection. We extract features without feature normalization and compute nearest-neighbor distances with FAISS using squared ℓ_2 distance. The mode-window selector clips the empirical OOD distribution to the 1st to 99th percentile range, builds a histogram with the code-level auto-bin heuristic, and selects the examples closest to the modal bin center.

Generators. The main method uses Stable Diffusion 3 in image-to-image mode. The generation ablation replaces it with FLUX or with a no-generation re-population baseline. In the completed PASS and DiffusionDB runs we increase the SD3 batch size to 12 to fit the 24-hour Slurm limit.

Source data specifics. PASS and DiffusionDB use `train[:10000]` with pretraining split override `[1.0, 0.0, 0.0]`. This keeps the source data budget aligned across methods while avoiding unnecessary pretraining validation overhead. The long-tailed ImageNet-100, CIFAR-10, and CIFAR-100 regimes follow the completed job configurations used in the final sweep.

Seeds and reporting. All main claims are supported by three seed averages. Tables report mean $\pm$ standard deviation over those seeds. The exact table values are generated from the completed job logs by the local export scripts in `scripts/export_paper_results.py` and `scripts/generate_neurips_paper_tables.py`.

B Compute Resources and Runtime

Most completed experiments use one node with two NVIDIA A100 80GB GPUs and a 24-hour wall-time limit. The successful DINO ablation uses four A100 80GB GPUs after reducing the number of local crops from six to two. Across the successful experiments we report, the aggregate training time is approximately 3022 wall-clock hours, or about 6142 GPU hours. This excludes failed runs, timeout runs, cache migration jobs, and other debugging or pilot jobs, so the full project required more compute than the successful totals alone.

Representative mean wall-clock times over three seeds are as follows. On ImageNet-100-LT source pretraining, SimCLR takes 16.9 hours and BRIDGE takes 21.8 hours. On PASS, SimCLR takes 10.2 hours and BRIDGE takes 14.9 hours. On DiffusionDB, SimCLR takes 11.6 hours and BRIDGE takes 16.4 hours. The successful DINO ablation takes 16.3 hours on four GPUs in its final configuration.

C Assets and Terms of Use

We use existing datasets and pretrained generators under their original release terms.

Source datasets. ImageNet-100-LT is derived from ImageNet, whose official access terms restrict the images to non commercial research and educational use. PASS is released under Creative Commons Attribution 4.0. DiffusionDB is released under CC0-1.0. CIFAR-10, CIFAR-100, Stanford Cars, FGVC Aircraft, Oxford Flowers, and Oxford-IIIT Pets are used from their official academic benchmark releases. When an official page provides download terms instead of a standard software style license, we follow those original terms.

Generators. Stable Diffusion 3 Medium is used under the Stability AI Community License. FLUX.1-dev is used only in the ablation and is subject to the FLUX.1-dev Non-Commercial License. We do not redistribute model weights, third party datasets, or generated image corpora as part of this submission.

D Broader Impact

A method that improves representation quality under biased source data can reduce the need for manual relabeling and can make SSL more useful in domains where balanced data are difficult to collect. That is a positive outcome. At the same time, our method relies on large image generators and on web-sourced datasets. These carry familiar risks related to license compliance, privacy, and unintended memorization or harmful content. Our experiments stay within existing dataset and model release terms and do not release new generators or scraped image collections. Any future deployment would still require a task-specific review of fairness, privacy, and content safety.

E Full Results

Table 8: Full linear-probe baseline comparison on ImageNet-100-LT pretraining.

Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Balanced	65.3 ± 1.2	37.0 ± 0.8	8.0 ± 0.5	15.1 ± 0.7	22.4 ± 1.6	$\mathbf{31.2 \pm 2.0}$	39.0 ± 1.0	31.1 ± 1.1
Imbalanced	65.4 ± 4.2	38.6 ± 1.4	7.6 ± 1.1	15.3 ± 0.9	23.7 ± 1.0	28.4 ± 1.3	39.6 ± 0.3	31.2 ± 1.4
BRIDGE (ours)	$\mathbf{70.7 \pm 1.7}$	$\mathbf{43.3 \pm 0.6}$	$\mathbf{9.9 \pm 1.5}$	$\mathbf{16.4 \pm 0.9}$	$\mathbf{26.6 \pm 2.7}$	$\mathbf{31.2 \pm 3.7}$	$\mathbf{42.0 \pm 1.2}$	$\mathbf{34.3 \pm 1.8}$

Table 9: Full kNN baseline comparison on ImageNet-100-LT pretraining.

Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Balanced	63.6 ± 0.7	32.8 ± 0.8	4.4 ± 0.4	9.0 ± 0.4	22.3 ± 0.1	17.2 ± 0.1	34.0 ± 1.1	26.2 ± 0.5
Imbalanced	63.9 ± 0.2	33.2 ± 0.3	4.5 ± 0.6	9.0 ± 0.3	22.5 ± 0.3	16.8 ± 0.2	34.3 ± 0.3	26.3 ± 0.3
BRIDGE (ours)	$\mathbf{66.0 \pm 0.9}$	$\mathbf{35.0 \pm 0.9}$	$\mathbf{5.0 \pm 0.4}$	$\mathbf{9.8 \pm 0.2}$	$\mathbf{24.8 \pm 0.4}$	$\mathbf{18.0 \pm 0.5}$	$\mathbf{35.6 \pm 1.2}$	$\mathbf{27.7 \pm 0.6}$

Table 10: Full linear-probe transfer results across all source datasets and comparison methods.

Source	Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ImageNet-100-LT	SimCLR	73.9 ± 0.5	46.8 ± 0.4	13.7 ± 0.3	19.6 ± 0.2	35.9 ± 7.2	38.6 ± 1.5	50.4 ± 0.5	39.8 ± 1.5
	TS	72.5 ± 0.4	44.0 ± 0.5	12.5 ± 1.2	15.0 ± 0.1	35.7 ± 1.8	40.6 ± 2.2	$\mathbf{56.7 \pm 1.2}$	39.6 ± 1.1
	SDCLR	41.1 ± 1.7	18.0 ± 0.6	1.4 ± 0.3	2.3 ± 0.5	10.8 ± 1.7	10.4 ± 0.3	21.9 ± 0.6	15.1 ± 0.8
	BRIDGE (ours)	$\mathbf{75.7 \pm 0.2}$	$\mathbf{48.8 \pm 0.3}$	$\mathbf{16.1 \pm 1.0}$	$\mathbf{22.4 \pm 0.3}$	$\mathbf{36.4 \pm 2.4}$	$\mathbf{42.7 \pm 1.9}$	53.7 ± 0.6	$\mathbf{42.3 \pm 1.0}$
	BRIDGE+TS (ours)	73.2 ± 0.1	45.5 ± 0.4	9.5 ± 1.2	13.0 ± 0.7	26.4 ± 2.7	34.4 ± 1.7	47.6 ± 0.6	35.7 ± 1.0
	BRIDGE+SDCLR (ours)	41.5 ± 0.5	17.7 ± 0.4	1.3 ± 0.3	2.4 ± 0.7	12.1 ± 0.8	10.5 ± 1.9	20.7 ± 1.0	15.2 ± 0.8
CIFAR-10-LT	SimCLR	67.8 ± 0.2	38.0 ± 1.0	6.5 ± 0.4	16.0 ± 0.9	19.2 ± 1.6	24.0 ± 1.3	32.0 ± 0.7	29.1 ± 0.9
	TS	68.5 ± 1.3	36.0 ± 1.7	4.3 ± 0.2	10.3 ± 0.1	13.1 ± 2.0	19.7 ± 2.3	31.5 ± 1.2	26.2 ± 1.3
	SDCLR	39.6 ± 1.1	15.6 ± 0.2	1.1 ± 0.1	2.4 ± 0.9	12.0 ± 1.6	9.1 ± 1.8	16.5 ± 1.6	13.8 ± 1.1
	BRIDGE (ours)	70.7 ± 2.1	41.5 ± 0.6	$\mathbf{8.6 \pm 0.4}$	$\mathbf{17.0 \pm 0.9}$	$\mathbf{24.0 \pm 3.2}$	$\mathbf{29.4 \pm 2.1}$	$\mathbf{34.4 \pm 1.5}$	$\mathbf{32.2 \pm 1.6}$
	BRIDGE+TS (ours)	$\mathbf{73.1 \pm 1.0}$	$\mathbf{42.0 \pm 0.3}$	4.4 ± 0.8	12.4 ± 0.5	10.2 ± 1.6	23.2 ± 1.2	34.2 ± 1.1	28.5 ± 0.9
	BRIDGE+SDCLR (ours)	39.1 ± 0.2	15.3 ± 1.0	1.1 ± 0.3	2.1 ± 0.6	8.9 ± 2.6	9.5 ± 1.1	16.8 ± 0.7	13.3 ± 0.9
CIFAR-100-LT	SimCLR	59.0 ± 1.5	31.1 ± 1.6	4.1 ± 0.3	11.2 ± 1.0	12.6 ± 3.4	20.1 ± 1.0	28.7 ± 1.3	23.8 ± 1.4
	TS	56.3 ± 0.2	28.0 ± 0.1	2.8 ± 0.6	8.5 ± 0.2	15.5 ± 2.5	16.2 ± 3.1	26.2 ± 1.8	21.9 ± 1.2
	SDCLR	37.1 ± 0.8	15.7 ± 0.2	1.9 ± 0.1	2.4 ± 0.7	9.4 ± 1.0	9.9 ± 1.7	16.2 ± 0.3	13.2 ± 0.7
	BRIDGE (ours)	$\mathbf{66.8 \pm 3.1}$	$\mathbf{38.3 \pm 2.2}$	$\mathbf{6.2 \pm 1.8}$	$\mathbf{14.6 \pm 1.6}$	$\mathbf{19.8 \pm 0.4}$	$\mathbf{22.7 \pm 2.3}$	$\mathbf{32.6 \pm 1.4}$	$\mathbf{28.7 \pm 1.8}$
	BRIDGE+TS (ours)	66.4 ± 3.3	38.0 ± 3.7	4.2 ± 1.0	10.9 ± 0.7	17.2 ± 3.6	21.6 ± 3.3	31.4 ± 2.3	27.1 ± 2.6
	BRIDGE+SDCLR (ours)	39.1 ± 1.9	14.9 ± 0.9	0.9 ± 0.2	3.0 ± 0.4	10.9 ± 2.5	7.9 ± 0.7	16.0 ± 0.3	13.2 ± 1.0
PASS-10k	SimCLR	72.3 ± 0.2	45.5 ± 0.5	11.0 ± 0.2	17.5 ± 1.4	33.3 ± 1.3	33.3 ± 0.9	44.7 ± 0.9	36.8 ± 0.8
	TS	71.5 ± 0.2	43.5 ± 0.4	10.2 ± 1.2	17.2 ± 0.2	33.1 ± 2.1	$\mathbf{36.4 \pm 1.8}$	49.4 ± 0.4	37.3 ± 0.9
	SDCLR	42.2 ± 1.1	17.7 ± 0.8	1.8 ± 0.3	2.2 ± 0.8	10.6 ± 1.3	8.2 ± 1.6	20.8 ± 0.4	14.8 ± 0.9
	BRIDGE (ours)	$\mathbf{74.2 \pm 0.2}$	$\mathbf{47.4 \pm 0.7}$	$\mathbf{13.0 \pm 1.2}$	$\mathbf{20.4 \pm 0.4}$	$\mathbf{39.4 \pm 3.4}$	35.8 ± 0.1	49.5 ± 1.0	$\mathbf{40.0 \pm 1.0}$
	BRIDGE+TS (ours)	69.1 ± 0.4	40.5 ± 0.8	12.1 ± 0.5	14.3 ± 0.3	32.7 ± 4.1	30.8 ± 1.0	$\mathbf{52.6 \pm 0.5}$	36.0 ± 1.1
	BRIDGE+SDCLR (ours)	45.5 ± 1.0	19.3 ± 0.2	1.5 ± 0.7	3.3 ± 0.2	11.0 ± 1.6	11.3 ± 1.2	20.9 ± 0.9	16.1 ± 0.8
DiffusionDB-10k	SimCLR	74.4 ± 0.5	48.6 ± 0.2	10.9 ± 0.7	17.6 ± 0.8	31.2 ± 2.4	36.1 ± 1.3	45.3 ± 0.5	37.7 ± 0.9
	TS	72.3 ± 0.3	44.4 ± 0.3	10.7 ± 0.6	15.4 ± 0.6	33.1 ± 1.1	$\mathbf{39.9 \pm 2.7}$	48.8 ± 0.2	37.8 ± 0.8
	SDCLR	42.4 ± 0.2	18.4 ± 0.2	1.1 ± 0.2	2.4 ± 0.1	13.0 ± 3.7	8.2 ± 1.4	21.0 ± 0.6	15.2 ± 0.9
	BRIDGE (ours)	$\mathbf{76.2 \pm 0.1}$	$\mathbf{49.8 \pm 0.1}$	$\mathbf{12.4 \pm 1.0}$	$\mathbf{20.4 \pm 0.2}$	$\mathbf{36.1 \pm 2.7}$	38.0 ± 3.2	48.9 ± 0.8	$\mathbf{40.3 \pm 1.2}$
	BRIDGE+TS (ours)	68.8 ± 0.1	40.2 ± 0.4	9.2 ± 1.1	12.8 ± 0.5	31.2 ± 2.4	34.8 ± 1.6	$\mathbf{51.0 \pm 0.6}$	35.4 ± 1.0
	BRIDGE+SDCLR (ours)	42.2 ± 0.7	17.6 ± 0.3	1.1 ± 0.6	2.4 ± 0.1	11.9 ± 3.6	11.7 ± 1.3	21.3 ± 0.8	15.5 ± 1.1

Table 11: Full kNN transfer results across all source datasets and comparison methods.

Source	Method	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ImageNet-100-LT	SimCLR	68.6 ± 0.4	37.3 ± 0.2	6.0 ± 0.7	11.2 ± 0.4	32.2 ± 0.7	23.6 ± 0.3	43.9 ± 0.9	31.8 ± 0.5
	TS	68.4 ± 0.3	36.5 ± 0.1	4.7 ± 0.2	8.3 ± 0.5	26.6 ± 1.0	23.6 ± 1.7	$\mathbf{52.2 \pm 0.5}$	31.5 ± 0.6
	SDCLR	32.5 ± 0.4	10.9 ± 0.2	1.0 ± 0.3	2.1 ± 0.1	10.7 ± 0.7	4.7 ± 0.2	11.2 ± 0.2	10.4 ± 0.3
	BRIDGE (ours)	$\mathbf{68.9 \pm 0.5}$	$\mathbf{37.8 \pm 0.1}$	$\mathbf{6.9 \pm 0.1}$	$\mathbf{12.1 \pm 0.2}$	$\mathbf{36.0 \pm 0.1}$	$\mathbf{25.3 \pm 0.9}$	48.2 ± 0.4	$\mathbf{33.6 \pm 0.3}$
	BRIDGE+TS (ours)	65.2 ± 0.2	33.7 ± 0.5	2.9 ± 0.4	6.6 ± 0.2	18.6 ± 0.4	15.8 ± 1.0	40.5 ± 0.8	26.2 ± 0.5
	BRIDGE+SDCLR (ours)	33.8 ± 0.4	11.5 ± 0.5	1.3 ± 0.4	2.0 ± 0.1	10.2 ± 0.7	4.9 ± 0.2	12.6 ± 0.6	10.9 ± 0.4
CIFAR-10-LT	SimCLR	66.9 ± 0.4	33.4 ± 0.3	3.7 ± 0.4	9.8 ± 0.7	16.5 ± 0.8	13.9 ± 0.8	24.0 ± 0.8	24.0 ± 0.6
	TS	66.3 ± 0.6	27.8 ± 0.4	2.4 ± 0.6	6.4 ± 0.2	10.5 ± 0.7	11.1 ± 0.4	22.0 ± 0.9	20.9 ± 0.5
	SDCLR	30.1 ± 0.3	10.0 ± 0.3	1.7 ± 0.3	2.2 ± 0.2	10.1 ± 0.7	4.4 ± 0.4	9.8 ± 0.3	9.7 ± 0.4
	BRIDGE (ours)	$\mathbf{69.0 \pm 1.2}$	$\mathbf{35.2 \pm 0.9}$	$\mathbf{4.2 \pm 0.6}$	$\mathbf{10.4 \pm 0.5}$	$\mathbf{17.0 \pm 0.6}$	$\mathbf{15.2 \pm 0.8}$	$\mathbf{25.2 \pm 0.5}$	$\mathbf{25.2 \pm 0.7}$
	BRIDGE+TS (ours)	67.7 ± 0.8	28.0 ± 0.8	2.1 ± 0.2	5.1 ± 0.7	8.1 ± 1.0	9.1 ± 0.8	21.8 ± 0.5	20.3 ± 0.7
	BRIDGE+SDCLR (ours)	31.3 ± 1.2	10.4 ± 0.5	1.0 ± 0.1	1.8 ± 0.3	9.6 ± 1.3	4.5 ± 0.1	10.1 ± 0.4	9.8 ± 0.6
CIFAR-100-LT	SimCLR	59.3 ± 0.8	30.1 ± 0.9	3.0 ± 0.3	7.4 ± 0.4	13.7 ± 0.2	10.8 ± 0.4	21.8 ± 1.1	20.9 ± 0.6
	TS	53.1 ± 0.6	24.6 ± 0.4	2.4 ± 0.4	5.5 ± 0.3	9.3 ± 0.4	8.3 ± 0.2	18.7 ± 1.0	17.4 ± 0.5
	SDCLR	25.7 ± 1.2	8.5 ± 0.6	0.9 ± 0.4	1.9 ± 0.2	7.3 ± 0.3	3.9 ± 0.3	8.3 ± 1.1	8.1 ± 0.6
	BRIDGE (ours)	$\mathbf{62.5 \pm 0.8}$	$\mathbf{32.5 \pm 0.3}$	$\mathbf{3.7 \pm 0.3}$	$\mathbf{8.7 \pm 0.5}$	$\mathbf{14.1 \pm 0.9}$	$\mathbf{12.1 \pm 0.7}$	$\mathbf{23.0 \pm 1.2}$	$\mathbf{22.4 \pm 0.7}$
	BRIDGE+TS (ours)	56.3 ± 1.2	27.8 ± 0.5	1.9 ± 0.0	4.8 ± 0.6	8.0 ± 1.0	9.0 ± 0.6	20.8 ± 0.8	18.4 ± 0.7
	BRIDGE+SDCLR (ours)	29.7 ± 2.2	9.8 ± 1.5	1.4 ± 0.4	2.3 ± 0.2	9.2 ± 1.1	4.8 ± 0.1	8.9 ± 0.6	9.4 ± 0.9
PASS-10k	SimCLR	67.2 ± 0.2	37.0 ± 0.3	$\mathbf{5.4 \pm 0.3}$	10.3 ± 0.5	30.4 ± 0.0	16.7 ± 0.5	38.3 ± 0.3	29.3 ± 0.3
	TS	65.2 ± 0.7	34.6 ± 0.3	4.4 ± 0.2	7.9 ± 0.2	24.5 ± 0.1	14.4 ± 0.4	39.3 ± 0.4	27.2 ± 0.3
	SDCLR	32.4 ± 1.5	11.5 ± 0.8	1.3 ± 0.2	2.0 ± 0.1	8.6 ± 0.7	5.2 ± 0.2	11.1 ± 0.8	10.3 ± 0.6
	BRIDGE (ours)	$\mathbf{67.8 \pm 0.6}$	$\mathbf{37.1 \pm 0.5}$	5.2 ± 0.4	$\mathbf{10.8 \pm 0.3}$	$\mathbf{33.7 \pm 0.6}$	$\mathbf{17.9 \pm 0.3}$	$\mathbf{41.9 \pm 0.4}$	$\mathbf{30.6 \pm 0.5}$
	BRIDGE+TS (ours)	59.0 ± 0.4	27.8 ± 0.5	3.3 ± 0.4	6.6 ± 0.0	23.3 ± 0.4	13.9 ± 0.3	39.2 ± 0.3	24.7 ± 0.3
	BRIDGE+SDCLR (ours)	37.9 ± 0.7	13.6 ± 0.0	1.2 ± 0.2	2.7 ± 0.1	8.9 ± 1.3	6.3 ± 0.1	13.9 ± 1.1	12.1 ± 0.5
DiffusionDB-10k	SimCLR	67.3 ± 0.1	36.2 ± 0.4	4.3 ± 0.2	8.8 ± 0.1	26.8 ± 0.4	17.3 ± 0.1	36.7 ± 0.8	28.2 ± 0.3
	TS	64.7 ± 0.4	33.6 ± 0.3	3.1 ± 0.5	7.3 ± 0.3	23.2 ± 0.4	17.0 ± 0.0	37.8 ± 0.6	26.7 ± 0.4
	SDCLR	32.6 ± 1.2	10.0 ± 0.3	1.3 ± 0.1	2.2 ± 0.2	6.1 ± 0.7	4.8 ± 0.4	10.6 ± 0.5	9.7 ± 0.5
	BRIDGE (ours)	$\mathbf{68.1 \pm 0.3}$	$\mathbf{37.0 \pm 0.2}$	$\mathbf{5.4 \pm 0.5}$	$\mathbf{9.5 \pm 0.1}$	$\mathbf{30.4 \pm 0.4}$	$\mathbf{18.5 \pm 0.1}$	$\mathbf{39.9 \pm 0.6}$	$\mathbf{29.8 \pm 0.3}$
	BRIDGE+TS (ours)	57.2 ± 0.7	25.6 ± 1.0	2.8 ± 0.3	6.8 ± 0.1	22.3 ± 0.5	15.4 ± 0.6	37.3 ± 0.6	23.9 ± 0.6
	BRIDGE+SDCLR (ours)	37.1 ± 1.3	11.8 ± 0.9	1.3 ± 0.1	2.2 ± 0.2	6.8 ± 0.5	5.9 ± 0.7	12.5 ± 0.7	11.1 ± 0.6

Table 12: Pretraining objective ablation with full linear-probe results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
SimCLR	$\mathbf{71.8 \pm 1.2}$	$\mathbf{43.5 \pm 0.5}$	$\mathbf{9.0 \pm 0.5}$	$\mathbf{16.9 \pm 0.3}$	$\mathbf{27.2 \pm 1.5}$	$\mathbf{33.2 \pm 3.3}$	$\mathbf{41.7 \pm 1.3}$	$\mathbf{34.7 \pm 1.2}$
MoCo	46.6 ± 3.1	18.4 ± 2.0	3.1 ± 0.5	4.3 ± 0.8	10.4 ± 3.2	15.0 ± 1.8	26.6 ± 1.9	17.8 ± 1.9
DINO	43.6 ± 0.7	17.8 ± 0.6	1.3 ± 0.5	1.6 ± 0.2	12.6 ± 4.4	8.7 ± 1.6	20.6 ± 1.1	15.2 ± 1.3

Table 13: Pretraining objective ablation with full kNN results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
SimCLR	$\mathbf{66.3 \pm 0.8}$	$\mathbf{35.3 \pm 0.4}$	$\mathbf{5.1 \pm 0.7}$	$\mathbf{9.5 \pm 0.1}$	$\mathbf{24.8 \pm 0.6}$	$\mathbf{18.9 \pm 1.0}$	$\mathbf{36.7 \pm 0.7}$	$\mathbf{28.1 \pm 0.6}$
MoCo	56.4 ± 1.6	25.9 ± 0.8	3.2 ± 0.4	5.2 ± 0.4	16.3 ± 1.1	12.2 ± 0.7	24.3 ± 0.7	20.5 ± 0.8
DINO	34.4 ± 0.2	12.4 ± 0.1	1.6 ± 0.1	2.4 ± 0.1	6.2 ± 0.8	5.4 ± 0.4	10.7 ± 0.8	10.4 ± 0.4

Table 14: Augmentation mechanism ablation with full linear-probe results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Stable Diffusion 3	$\mathbf{75.0 \pm 0.2}$	$\mathbf{48.4 \pm 0.2}$	$\mathbf{15.1 \pm 1.2}$	$\mathbf{20.6 \pm 0.7}$	$\mathbf{37.4 \pm 2.0}$	$\mathbf{41.2 \pm 0.7}$	$\mathbf{51.8 \pm 1.1}$	$\mathbf{41.4 \pm 0.8}$
FLUX	70.7 ± 0.4	42.1 ± 1.4	10.3 ± 1.3	17.5 ± 0.9	29.4 ± 4.6	31.5 ± 2.9	43.7 ± 1.4	35.0 ± 1.9
No-generation re-population	67.0 ± 2.6	39.4 ± 2.2	8.0 ± 0.6	15.4 ± 0.5	25.7 ± 1.4	31.7 ± 3.8	40.6 ± 0.5	32.6 ± 1.7

Table 15: Augmentation mechanism ablation with full kNN results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Stable Diffusion 3	$\mathbf{68.3 \pm 0.4}$	$\mathbf{36.8 \pm 0.2}$	$\mathbf{6.5 \pm 0.4}$	$\mathbf{11.7 \pm 0.6}$	$\mathbf{35.1 \pm 0.4}$	$\mathbf{24.2 \pm 0.3}$	$\mathbf{45.9 \pm 0.5}$	$\mathbf{32.7 \pm 0.4}$
FLUX	65.8 ± 1.0	35.4 ± 0.8	5.7 ± 0.7	9.4 ± 0.9	27.7 ± 1.0	18.8 ± 1.3	38.7 ± 1.2	28.8 ± 1.0
No-generation re-population	64.5 ± 0.1	33.4 ± 0.6	4.2 ± 0.6	9.0 ± 0.4	23.0 ± 0.5	17.0 ± 0.4	34.2 ± 1.2	26.5 ± 0.5

Table 16: Selection rule ablation with full linear-probe results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Mode-window	**71.9 ± 0.3**	43.4 ± 0.7	9.0 ± 0.5	16.9 ± 0.2	27.9 ± 3.1	31.6 ± 3.3	41.7 ± 1.3	34.6 ± 1.3
Top-tail	70.8 ± 0.5	43.1 ± 1.0	**10.0 ± 1.7**	15.9 ± 1.3	27.7 ± 1.0	30.5 ± 1.2	**42.2 ± 1.4**	34.3 ± 1.2
Uniform	71.5 ± 0.2	**43.8 ± 0.7**	9.3 ± 0.9	**17.5 ± 1.2**	**32.9 ± 0.4**	**31.7 ± 3.9**	42.0 ± 0.9	**35.5 ± 1.2**

Table 17: Selection rule ablation with full kNN results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
Mode-window	**66.3 ± 0.8**	35.3 ± 0.4	**5.1 ± 0.7**	9.5 ± 0.1	24.8 ± 0.6	18.9 ± 1.0	36.7 ± 0.7	28.1 ± 0.6
Top-tail	65.7 ± 0.8	34.4 ± 0.2	4.1 ± 0.4	9.6 ± 0.2	23.9 ± 0.9	18.1 ± 0.8	35.4 ± 1.2	27.3 ± 0.6
Uniform	66.1 ± 1.0	**35.7 ± 0.3**	4.9 ± 0.1	**10.2 ± 0.7**	**25.3 ± 0.4**	**19.0 ± 0.8**	**37.4 ± 0.6**	**28.4 ± 0.6**

Table 18: Cycle count ablation with full linear-probe results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
2 cycles	67.9 ± 0.6	40.7 ± 0.6	9.4 ± 1.8	16.0 ± 1.5	**30.5 ± 3.2**	27.4 ± 1.7	40.1 ± 0.4	33.1 ± 1.4
5 cycles	70.7 ± 1.7	43.3 ± 0.6	**9.9 ± 1.5**	16.4 ± 0.9	26.6 ± 2.7	31.2 ± 3.7	42.0 ± 1.2	34.3 ± 1.8
10 cycles	70.5 ± 1.2	43.8 ± 0.6	9.2 ± 0.9	17.2 ± 1.6	24.8 ± 5.2	31.3 ± 1.7	41.9 ± 0.6	34.1 ± 1.7
20 cycles	**71.6 ± 1.1**	**44.6 ± 0.9**	9.8 ± 0.3	**17.3 ± 1.2**	27.0 ± 2.1	**33.3 ± 4.4**	**42.1 ± 1.5**	**35.1 ± 1.6**

Table 19: Cycle count ablation with full kNN results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
2 cycles	65.1 ± 0.1	34.1 ± 0.5	4.6 ± 0.3	9.3 ± 0.3	23.8 ± 0.3	17.3 ± 0.7	34.5 ± 0.6	26.9 ± 0.4
5 cycles	66.0 ± 0.9	35.0 ± 0.9	**5.0 ± 0.4**	9.8 ± 0.2	24.8 ± 0.4	18.0 ± 0.5	35.6 ± 1.2	27.7 ± 0.6
10 cycles	66.4 ± 0.5	35.4 ± 0.3	4.8 ± 0.3	**10.4 ± 0.6**	**25.2 ± 0.8**	**18.6 ± 0.9**	36.0 ± 0.7	**28.1 ± 0.6**
20 cycles	**66.5 ± 0.7**	**35.6 ± 0.8**	4.8 ± 0.3	9.5 ± 0.4	24.9 ± 0.5	18.6 ± 1.2	**36.3 ± 1.2**	28.0 ± 0.7

Table 20: Backbone architecture ablation with full linear-probe results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ResNet-18	67.7 ± 0.8	40.4 ± 0.5	**10.6 ± 0.7**	15.4 ± 0.3	26.1 ± 2.4	29.5 ± 2.7	41.0 ± 1.2	32.9 ± 1.2
ResNet-50	**71.7 ± 1.0**	**44.0 ± 1.1**	9.0 ± 0.5	**17.4 ± 0.7**	28.8 ± 1.7	**32.0 ± 1.6**	41.7 ± 1.3	**34.9 ± 1.1**
ViT-S	62.7 ± 0.4	37.6 ± 0.6	8.8 ± 0.5	12.2 ± 0.2	35.1 ± 4.2	30.2 ± 1.6	42.5 ± 1.2	32.7 ± 1.3
ViT-B	66.0 ± 2.3	40.6 ± 2.3	8.3 ± 0.5	14.3 ± 0.8	**38.8 ± 2.6**	31.3 ± 2.1	**44.5 ± 1.7**	34.8 ± 1.8

Table 21: Backbone architecture ablation with full kNN results.

Setting	C10	C100	Cars	Aircraft	Flowers	Pets	IN100-LT	Avg.
ResNet-18	65.0 ± 0.3	34.4 ± 0.3	4.7 ± 0.3	**9.6 ± 0.4**	23.5 ± 1.4	16.6 ± 0.3	36.3 ± 1.1	27.2 ± 0.6
ResNet-50	66.3 ± 0.8	35.3 ± 0.4	5.1 ± 0.7	9.5 ± 0.1	24.8 ± 0.6	18.9 ± 1.0	36.7 ± 0.7	28.1 ± 0.6
ViT-S	67.3 ± 0.4	37.5 ± 0.3	5.3 ± 0.2	7.6 ± 0.1	36.2 ± 0.7	20.1 ± 1.0	42.1 ± 1.0	30.9 ± 0.5
ViT-B	**69.3 ± 0.8**	**40.1 ± 0.8**	**6.0 ± 0.2**	8.3 ± 0.2	**39.8 ± 0.8**	**21.9 ± 1.4**	**45.1 ± 0.7**	**32.9 ± 0.7**

F Algorithm

Algorithm 1 BRIDGE training loop with mode-window selection

```
Initialize source set $D^{(0)}$, encoder $f_{\theta^{(0)}}$, frozen generator $g_\phi$, cycle budgets $\{E_c\}_{c=0}^{C-1}$
for $c = 0, \dots, C-1$ do
    Train or continue training $f_{\theta^{(c)}}$ on $D^{(c)}$ for $E_c$ epochs
    Compute embeddings $Z^{(c)} = \{f_{\theta^{(c)}}(x_i) : x_i \in D^{(c)}\}$
    Compute mean $k$NN scores $\{d_i^{(c)}\}_{i=1}^{N^{(c)}}$ from $Z^{(c)}$
    Clip $\{d_i^{(c)}\}$ to the 1st to 99th percentile range and build a histogram
    Let $m^{(c)}$ be the center of the modal histogram bin
    Select $S^{(c)} \subseteq [N^{(c)}]$ with $|S^{(c)}| = B$ whose scores are closest to $m^{(c)}$
    Form $\widetilde{D}^{(c)} = \{g_\phi(x_i, \epsilon_j) \mid i \in S^{(c)},\, j \in [G]\}$
    Update $D^{(c+1)} = D^{(c)} \cup \widetilde{D}^{(c)}$
end for
Return the final encoder and repaired source set
```

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